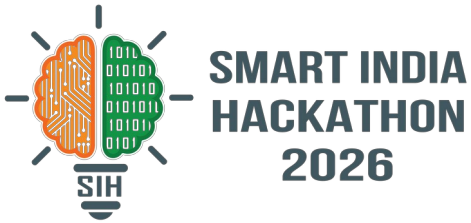


SMART INDIA HACKATHON 2026

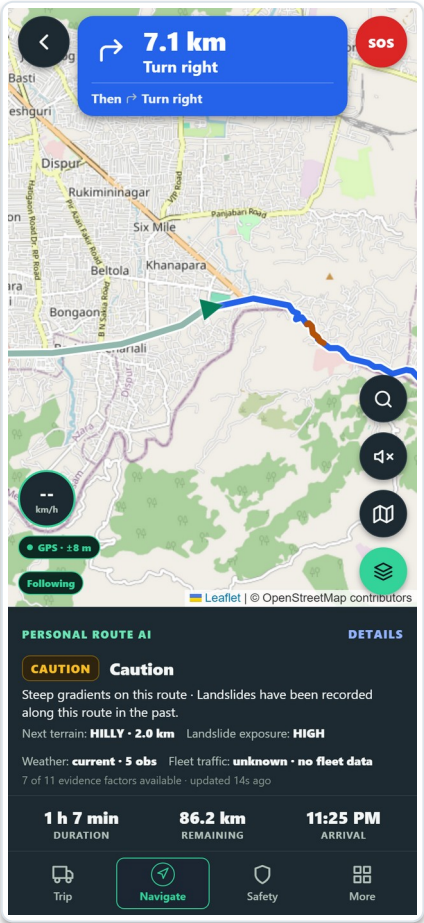
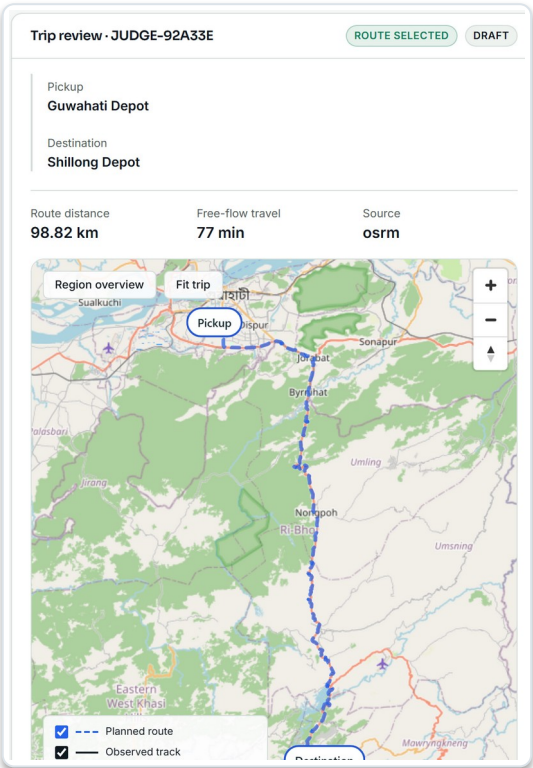


RASTA AI

Route Intelligence for Essential-Supply Logistics in North Eastern India

THE ASK	SIH26002		PROBLEM STATEMENT ID
			MDoNER · Smart Automation · Software
THE RECORD	PROBLEM STATEMENT TITLE	AI-Based Smart Logistics and Accessibility Intelligence Platform for North Eastern Region (NER)	
	THEME	Smart Automation	
	PS CATEGORY	Software	
	TEAM ID	17 (internal group no.)	
	TEAM NAME	NER-AI LOGISTICS	
THE POINT	SHORTEST IS NOT ALWAYS USABLE.		
	“Can this essential-supply truck reliably use this corridor now?”		
	End-to-end working prototype · physical-phone validated · live demo corridor Guwahati → Shillong		

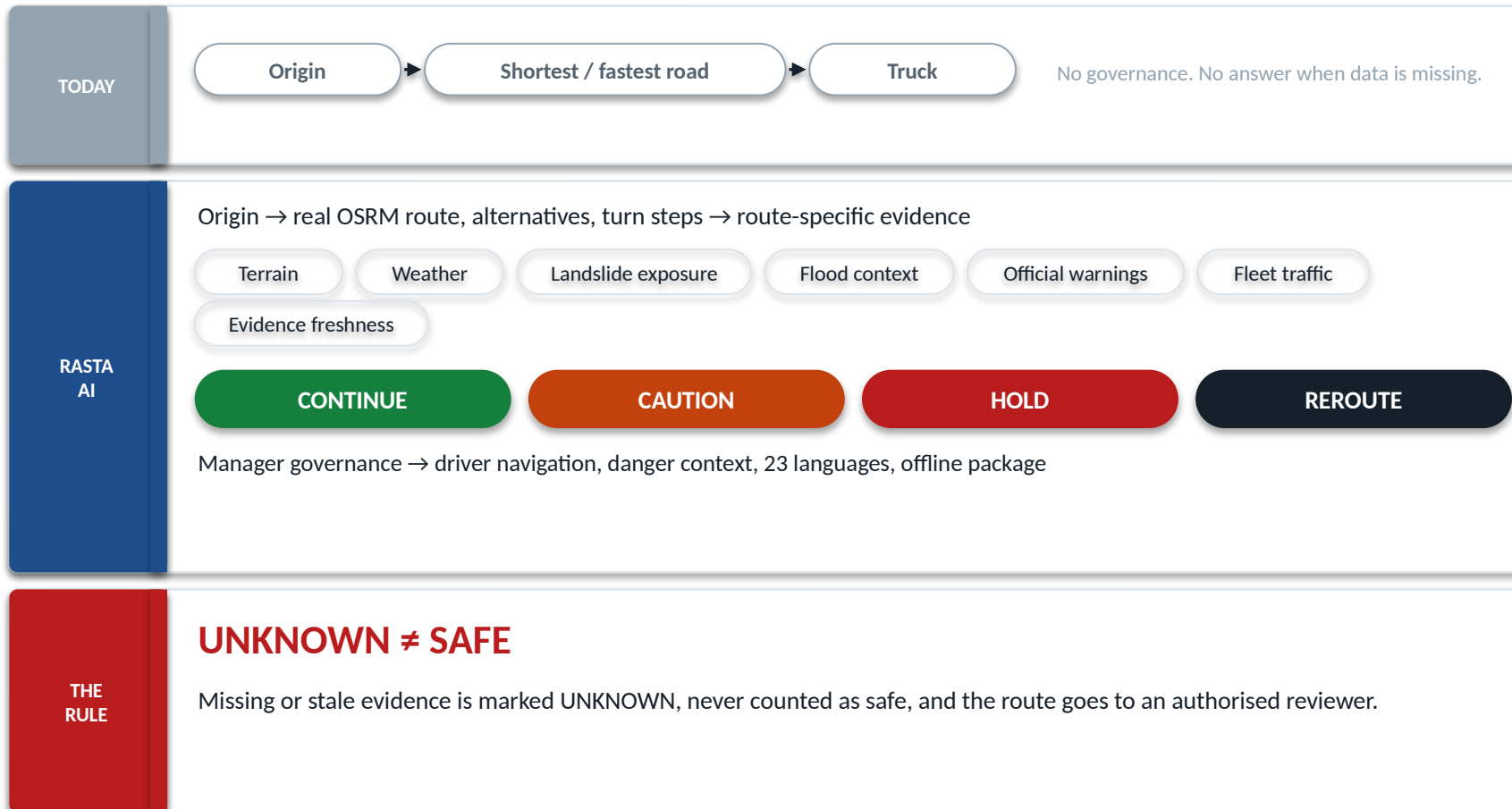
LIVE WORKING SYSTEM



Manager route review · Driver navigation — real screens

Not the shortest road. **The road that is operationally usable now.**

MANAGER • CHECK CONDITIONS

**REQUIRES REVIEW**

Hazard evidence is incomplete or elevated. An authorised reviewer must review this route before selection. Refresh conditions after review.

Landslide data is not available - no source is connected. · Steep gradients on this route · Landslides have been recorded along this route in the past. · The landslide inventory ends in 2017 - recent years are not covered. · River levels near this route are close to their recent average. · No official disaster alert names a district on this route right now. · No fleet truck has driven this road in the last fifteen minutes, so traffic is unknown.

Unavailable: Fuel range, Landslide, Road quality, Truck restrictions

Evidence coverage · Weather AVAILABLE · Terrain AVAILABLE · Warnings AVAILABLE · Traffic UNKNOWN

TERRAIN**0.5 KM STEEP**

49–1436 m · 1975 m climb · steepest 10.3%
flat 61.8 km · rolling 29.6 km · hilly 7.0 km · steep 0.5 km
Copernicus DEM GLO-90 via Open-Meteo + SRTM 30 m via OpenTopoData · 100% of 198 samples answered

LANDSLIDE HISTORY**HIGH EXPOSURE**

22 recorded landslides within 5 km of the road, nearest 0.4 km
6 more nearby placed too imprecisely to count · HIGH is three or more; MODERATE is one or two
NASA Global Landslide Catalog · inventory 2007–2017 — aged, recent years not covered

RIVER LEVELS**NORMAL**

Discharge at 5 river cells along the corridor, highest 1.0× its own 30-day mean · GloFAS via Open-Meteo, 2026-09-14. A level, not a flood forecast and not a road-closure claim.

OFFICIAL ALERTS**NONE ON CORRIDOR**

NDMA SACHET CAP feed, 99 alerts nationwide at 10:13:54 PM · placed by district name (Dispur, East Khasi Hills, Kamrup Metropolitan, Myllem, Ri-Bhoi, Umling, Umsning) · 4 more active elsewhere in the corridor states, not placeable on this road.

FLEET TRAFFIC**UNKNOWN**

Real screen · evidence · freshness · UNKNOWN

TECHNICAL APPROACH

**VERIFIED
DATA****WEATHER**

Open-Meteo · MET Norway

TERRAINCopernicus DEM ·
OpenTopoData**WARNINGS**

NDMA SACHET (CAP)

FLOOD

GloFAS

LANDSLIDE HISTORY

NASA GLC 2007–17

TRAFFIC

fleet GPS fixes

DECISION**ROUTE-SPECIFIC EVIDENCE · FastAPI backend**

every factor sampled along the selected road · PostgreSQL + PostGIS (Supabase) · OSRM routes, alternatives, turn steps

DETERMINISTIC SAFETY POLICY

11 factors · reason codes · UNKNOWN ≠ SAFE · reviewer authorisation when hazard data is missing

CONTINUE**CAUTION****HOLD****REROUTE**

MANAGER REVIEW

DRIVER NAVIGATION

ROUTE-AHEAD 60 s

CONDITIONS CHANGE?

REASSESS

**BUILT
WITH****CLIENTS** Manager console — React + TypeScript, manager accounts only · Driver app — React Native / Expo, Android; one login, server-decided role, offline trip package**STACK** FastAPI · PostgreSQL / PostGIS (Supabase) · React · React Native / Expo · OSRM · OpenStreetMap · Gemini**EVIDENCE** Open-Meteo / MET Norway · NDMA SACHET · GloFAS · NASA historical landslides · OpenTopoData / Copernicus DEM
Re-scored every 60 s along the road ahead; a reroute is proposed to the manager, never applied silently.**WHO DECIDES****DETERMINISTIC SAFETY ENGINE**

makes every operational decision

GEMINI / OPENROUTER (LLM)

explains the decision in the driver's language — never decides

EXPERIMENTAL ML

research channel only; held outside routing by the safety gate

GOVERNANCE CHAIN

Verified evidence



Deterministic safety policy



Manager governance



Driver action

Not a concept. A working end-to-end prototype.

LIFECYCLE

PLAN

REVIEW

DISPATCH

ACCEPT

TRUCK VERIFY

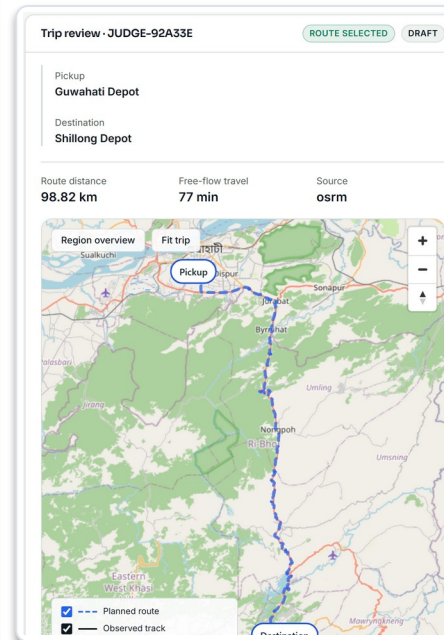
NAVIGATE

OFF-ROUTE

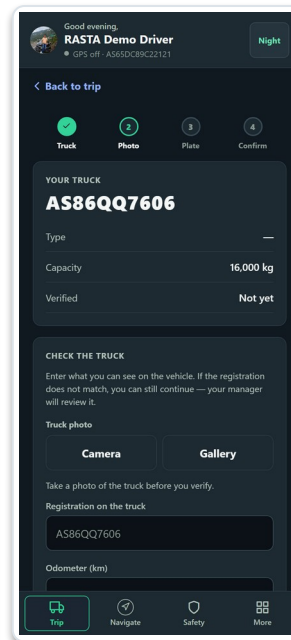
MANAGER REROUTE APPROVAL

DELIVERY

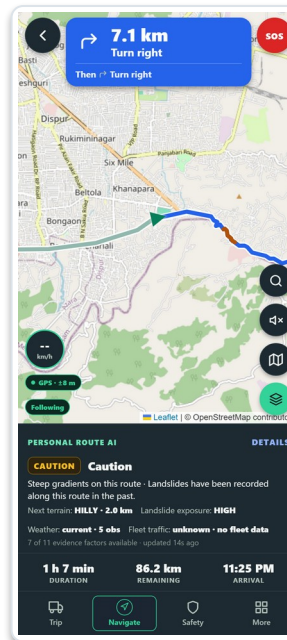
EVIDENCE



Manager · route review



Driver · truck check



Driver · navigation

The whole lifecycle above was run end to end on a physical Android phone against the hosted backend, on the real road.

12/12 physical judge-flow steps

9/9 physical role-flow steps

1138+ backend · 621+ driver · 170+ manager tests

PHYSICAL ANDROID · CERTIFIED

Canonical demo

Guwahati → Shillong · ≈ 98.8 km real route

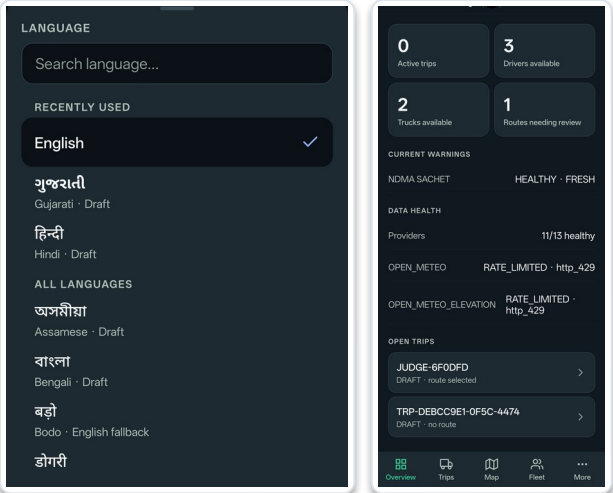
KNOWN LIMITS → MITIGATION Render cold start → warm /health before the demo · provider rate limits → cached evidence + explicit UNKNOWN · background push needs Firebase config → in-app alerts already work · experimental ML → outside routing until the gate passes

IMPACT AND BENEFITS

RASTA AI does not ask only “Which road is shortest?” It asks “Can this truck reliably use this corridor now?”

WHO GAINS	FLEET MANAGERS	DRIVERS	REMOTE COMMUNITIES	GOVERNMENT / OPERATIONS
	Route usability and uncertainty visible before dispatch; auditable decisions.	Safer navigation, danger context, emergency support, 23 languages.	More resilient access to food, medicine, fuel and relief supplies.	Auditable route evidence and disruption visibility per corridor.
PROVEN	23 selectable languages, explicit fallback status	11 route factors in every decision	12/12 judge-flow steps certified on a physical phone	100 % of reroutes reviewed by a human
	60 s route-ahead monitor while driving		0 unknown factors counted as safe	
GOVERNED	FOOD • MEDICINE • FUEL • RELIEF SUPPLIES — essential logistics for hill communities, decided on evidence			
	Verified evidence → Deterministic safety policy → Manager governance → Driver action No invented percentages: impact is stated as who gains what, backed by the working system.			

REAL SCREENS • APK 1.0.18



Driver • 23 languages

Manager • mobile

Real screens, APK 1.0.18. One login, server-decided role: a manager gets a mobile fleet view, a driver gets navigation. Each language shows its status — Verified, Draft or English fallback — so nobody is misled.

RESEARCH AND REFERENCES

SOURCES

ROUTING / PLACES

OSRM • OpenStreetMap • Nominatim

real road geometry, alternatives and turn steps; place search on the corridor

OFFICIAL / FLOOD

NDMA SACHET (CAP) • Copernicus GloFAS

official warnings polled with freshness; river discharge against its 30-day mean

AI ASSISTANCE

Google Gemini

explains the decision in the driver's language — never safety authority

WEATHER / TERRAIN

Open-Meteo • MET Norway • Copernicus DEM • OpenTopoData

forecast sampled along the road; climb, gradient and steep segments (GLO-90)

LANDSLIDE HISTORY

NASA Global Landslide Catalog

historical exposure; inventory 2007–2017, ≤5 km location-accuracy filter

PROBLEM STATEMENT

MDoNER • SIH26002

AI-based smart logistics and accessibility intelligence for the NER

EXPERIMENTAL LANDSLIDE RESEARCH

95 %

RECALL

on the NER geographic holdout — 154 catalogued events, no NER row used in training

SAFETY GATE

NOT DEPLOYED

FPR 50 % too high for production deployment

TRAIN

HELD-OUT NER TEST

USEFUL SIGNAL FOUND

FALSE POSITIVES TOO HIGH

SAFETY GATE

NOT DEPLOYED

VALIDATION BEFORE AUTOMATION. Strong recall was not enough. Held-out testing exposed the false-positive rate, so the gate keeps the model out of routing.

Precision 0.32 • F1 0.48 • PR-AUC 0.55 • ROC-AUC 0.83 • Brier 0.12 • calibrated logistic regression on NASA GLC + ERA5-Land rain + DEM slope • docs/HAZARD_ERROR_ANALYSIS.md